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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/566,821	12/04/2023	Kazumasa MATSUMOTO	PTB-427-559	2669

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EXAMINER
GOLDBERG, JEANINE ANNE

ART UNIT	PAPER NUMBER
1682	

NOTIFICATION DATE	DELIVERY MODE
05/29/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/566,821	Applicant(s) MATSUMOTO et al.	
	Examiner JEANINE A GOLDBERG	Art Unit 1682	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 5/18/26.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 13 and 15-18 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 13 and 15-18 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☑ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. This action is in response to the papers filed May 18, 2026. Currently, claims 13, 15-18 are pending.

3. All arguments have been thoroughly reviewed but are deemed non-persuasive for the reasons which follow. This action is made FINAL.

4. Any objections and rejections not reiterated below are hereby withdrawn.

Priority

5. This application claims priority to

JAPAN 2021-097456 06/10/2021
JAPAN 2021-201094 12/10/2021

It is noted that a translation of the foreign document has not been received.

Drawings

6. The drawings are acceptable.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

7. Claims 13, 15-18 are rejected under 35 U.S.C. 101 because the claimed invention is directed to non-statutory subject matter.

35 U.S.C. § 101 requires that to be patent-eligible, an invention (1) must be directed to one of the four statutory categories, and (2) must not be wholly directed to subject matter encompassing a judicially recognized exception. M.P.E.P. § 2106. Regarding judicial exceptions, "[p]henomena of nature, though just discovered, mental processes, and abstract intellectual concepts are not patentable, as they are the basic tools of scientific and technological work." *Gottschalk v. Benson*, 409 U.S. 63, 67 (1972); see also M.P.E.P. § 2106, part II.

Based upon consideration of the claims as a whole, as well as consideration of elements/steps recited in addition to the judicial exception, the present claims fail to meet the elements required for patent eligibility.

Question 1

The claimed invention is directed to a process that involves a natural principle and a judicial exception.

Question 2A Prong I

The claims are taken to be directed to an abstract idea, a law of nature and a natural phenomenon.

Claim 13 is directed to "a method for treating cancer" by measuring expression level of SPSB2 wherein the measured expression is **higher than that of a control** and excising a cancer tissue or performing anti-cancer agent treatment to the subject.

Claim 13 is directed to a process that involves the judicial exceptions of an abstract idea (i.e. the abstract idea of "the expression level ...is higher than that of a control") and a law of nature/natural phenomenon (i.e. the natural correlation between the expression level and cancer).

The claims do not include additional elements that are sufficient to amount to significantly more than the judicial exception for the reasons that follow.

Claim 13 recites a comparison between the expression level and a normal control that is deemed an abstract idea (see MPEP 2106.04(a)(2)(III)(A); • claims to "comparing BRCA sequences and determining the existence of alterations," where the claims cover any way of comparing BRCA sequences such that the comparison steps can practically be performed in the human mind, *University of Utah Research Foundation v. Ambry Genetics,* 774 F.3d 755, 763, 113 USPQ2d 1241, 1246 (Fed. Cir. 2014)).

A correlation that preexists in the human is an unpatentable phenomenon. The association between expression level of SPSB2 and cancer is a law of nature/natural phenomenon. The indicates recitation which tells users of the process to predict cancer in the sample, amounts to no more than an "instruction to apply the natural law". This indication is no more than a mental step. Even if the step requires something more such as to verbalize the discovery of the natural law, this mere verbalization is not an application of the law of nature to a new and useful end. The indicates does not require the process user to do anything in light of the correlation. The indicates recitation fails

to provide the "practical assurance" sought by the *Prometheus Court* that the "process is more than a drafting effort designed to monopolize the law of nature itself."

Question 2A Prong II

The exception is not integrated into a practical application of the exception. The claims do not recite any additional elements that integrate the exception into a practical application of the exception. While the claim recites measuring expression level of SPSB2 and excising a cancer tissue or performing anti-cancer agent treatment, this is not an integration of the exception into a practical application. The measuring expression step is a data collection step.

The excising a cancer tissue or performing anti-cancer agent treatment is general in nature and not particularly drawn to the judicial exception. While the claim recites excising a cancer tissue or performing anti-cancer agent treatment, this is not an integration of the exception into a practical application. The limitation does not indicate how the patient is to be treated, or what the treatment is but instead covers any possible anti-cancer agent treatment that a doctor decides to administer to the patient. The excising a cancer tissue or performing anti-cancer agent treatment limitation fails to meaningfully limit the claim because it does not require any particular application of the recited calculation, and is at best the equivalent of merely adding the words "apply it" to the judicial exception.

Furthermore, the treatment step is optional. The claim provides the treatment is performed "only when the measured expression level is higher than a control". Since the claim is conditional and cancer tissue is only excised "when" measured expression level

is higher and a control the claim does not require the excising or administering the treatment and as such does not provide an integration of the judicial exception. Accordingly, the claims are directed to judicial exceptions.

Question 2B

The second step of Alice involves determining whether the remaining elements, either in isolation or combination with the other non patent ineligible elements, are sufficient to "'transform the nature of the claim' into a patent eligible application" *Alice*, 134 S. Ct. at 2355 (quoting *Mayo*, 132 S. Ct. at 1297).

The claims are not sufficiently defined to provide a method which is significantly more from a statement of a natural principle for at least these reasons:

The claims do not include applying the judicial exception, or by use of, a particular machine. The claims do not tie the steps to a "particular machine" and therefore do not meet the machine or transformation test on these grounds. The use of machines generally does not impose a meaningful limit on claim scope.

The claims also do not add a specific limitation other than what is well-understood, routine and conventional in the field. The measuring expression level is mere data gathering step that amounts to extra solution activity to the judicial exception. It merely tells the users of the method to determine the biomarkers of a sample without further specification as to how the sample should be analyzed. The claims do not recite a new, innovative method for such determination. The measuring expression

level essentially tells users to determine the expression through whatever known processes they wish to use.

The step of determining the expression levels was well known in the art at the time the invention was made. The prior art teaches that expression analysis using commercially available biochips and arrays that comprise the claimed genes. The steps are recited at a high level of generality. The claim merely instructs a scientist to use any expression analysis assay to determine the expression level status. The claim does not require the use of any particular non-conventional reagents. When recited at this high level of generality, there is no meaningful limitation that distinguishes this step from well understood, routine and conventional activities engaged in by scientists prior to applicant's invention and at the time the application was filed.

Additionally, the teachings in the specification demonstrate the well understood, routine, conventional nature of additional elements because it teaches that the additional elements were well known. Specifically, the specification teaches the expression level of the SPSB2 gene was examined using the Cancer Genome Atlas which is a public database (page 15, para 43).

Further it is noted that the courts have recognized the following laboratory techniques as well-understood, routine, conventional activity in the life science arts when they are claimed in a merely generic manner (e.g., at a high level of generality) or as insignificant extra-solution activity.

- Analyzing DNA to provide sequence information or detect allelic variants, Genetic Techs., 818 F.3d at 1377; 118 USPQ2d at 1546;
- Amplifying and sequencing nucleic acid sequences, University of Utah Research Foundation v. Ambry Genetics, 774 F.3d 755, 764, 113 USPQ2d 1241, 1247 (Fed. Cir. 2014)

For these reasons the claims are rejected under section 101 as being directed to non-statutory subject matter.

Response to Arguments

The response traverses the rejection. The response asserts the claim has been amended to specify the sample is a tissue sample and the cancer is bladder, pancreatic or hepatocellular cancer. The response also argues the claim includes concrete physical steps and not mere mental processes. This argument has been considered but is not convincing because the claim remains directed to a judicial exception. The claim does not require a particular treatment step and the concrete physical steps in addition to the judicial exception are routine and conventional. Thus for the reasons above and those already of record, the rejection is maintained.

Claim Rejections - 35 USC § 112-Scope of Enablement

The following is a quotation of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

8. Claims 13, 15-18 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, because the specification, while being enabling for methods of determining the expression level of SPSB2, does not reasonably provide enablement for a method of treating any cancer by measuring an expression level of SPSB2 in any sample. The specification does not enable any person skilled in the art to which it

pertains, or with which it is most nearly connected, to use the invention commensurate in scope with these claims.

Factors to be considered in determining whether a disclosure meets the enablement requirement of 35 USC 112, first paragraph, have been described by the court in *In re Wands*, 8 USPQ2d 1400 (CA FC 1988). *Wands* states at page 1404,

> "Factors to be considered in determining whether a disclosure would require undue experimentation have been summarized by the board in Ex parte Forman. They include (1) the quantity of experimentation necessary, (2) the amount of direction or guidance presented, (3) the presence or absence of working examples, (4) the nature of the invention, (5) the state of the prior art, (6) the relative skill of those in the art, (7) the predictability or unpredictability of the art, and (8) the breadth of the claims."

The nature of the invention and breadth of claims

The claims are drawn to a method of treating bladder, pancreatic or hepatocellular cancer by measuring an expression level of SPSB2 in any sample.

The invention is in a class of invention which the CAFC has characterized as "the unpredictable arts such as chemistry and biology." Mycogen Plant Sci., Inc. v. Monsanto Co., 243 F.3d 1316, 1330 (Fed. Cir. 2001).

The unpredictability of the art and the state of the prior art

The art teaches analysis of Spsb2 in hepatocellular carcinoma (see Tian et al. Genes, Vol. 16, No. 346, March 17, 2025). Tian teaches differential expression analysis of SPSB2 in pan-cancer and Liver cancer. Tian illustrates not all cancers overexpress SPSB2. As seen in Figure 2A, many of the comparison were not significant. Even more, KICH, KIRC, LUSC, PCPG, PRAD showed decreased expression.

Habermann et al. (US 2013/0196876, August 1, 2013) teaches analyzing SPSB2 protein by incubating a sample with antibodies with a protein encoded by SPSB2 (see

para 10) and comparing the amount to a control. Habermann teaches SPSB2 is downregulated compared to healthy controls in **pancreatic adenomas** (Table 1).

Guidance in the Specification.

The specification provides no evidence that the broad scope of the claims is enabled.

The specification teaches SPSB2 gene is not over expressed in all cancers. The specification teaches SPSB2 was strongly expressed in bladder and pancreatic cancer and moderately expressed in hepatocellular carcinoma and ovarian cancer, but not expressed in kidney, prostate, esophageal, gastric, colon, breast or lung cancers (para 60).

The specification teaches SPSB2 can not be detected in all types of samples. In particular, SPSB2 protein was not found in exosomes from serum or normal healthy patients (para 68).

The specification teaches urinary tract infections also show expression of SPSB2 in patients with urinary tract infection is higher than bladder cancer patients (para 75).

The guidance provided by the specification amounts to an invitation for the skilled artisan to try and follow the disclosed instructions to make and use the claimed invention.

Quantity of Experimentation

The quantity of experimentation in this area is extremely large since there is significant number of parameters which would have to be studied to enable the skilled artisan to practice the claimed invention as broadly as claimed.

The claims are directed to a method that requires measuring an expression level of SPSB2 and detecting overexpression of SPSB2 as indicative of bladder, pancreastic or hepatocellular cancer. Neither the specification nor the art

teaches that all cancers are associated with over expression of SPSB2. The specification teaches SPSB2 gene is not over expressed in all cancers. The specification teaches SPSB2 was strongly expressed in bladder and pancreatic cancer and moderately expressed in hepatocellular carcinoma and ovarian cancer, but not expressed in kidney, prostate, esophageal, gastric, colon, breast or lung cancers (para 60). Tian illustrates not all cancers overexpress SPSB2. As seen in Figure 2A, many of the comparison were not significant. Even more, KICH, KIRC, LUSC, PCPG, PRAD showed decreased expression. Habermann teaches SPSB2 is downregulated compared to healthy controls in **pancreatic adenomas** (Table 1). Therefore, it would require further unpredictable experimentation to determine which cancers overexpress SPSB2 and which cancers do not show an association or are under expressed. The art teaches pancreatic cancer is downregulated in **pancreatic adenomas** compared to healthy controls. This would require further unpredictable and undue experimentation to study the vast number of cancers encompassed by the claims.

The claims are directed to measuring expression level of SPSB in any biological sample including exosomes extracted from urine and urine. The specification teaches SPSB2 can not be detected in all types of samples. In particular, SPSB2 protein was not found in exosomes from serum or normal healthy patients (para 68). The skilled artisan would be unable to use exosome from serum to determine whether a subject has overexpression of SPSB2.

The claims are directed to using any control subject including subject that have with urinary tract infections. The specification teaches urinary tract infections also show expression of SPSB2 in patients with urinary tract infection is higher than bladder cancer patients (para 75). It would require further unpredictable and undue experimentation to determine how to diagnose cancer in patients with urinary tract infections using SPSB2 expression. This would require significant inventive effort, with

each of the many intervening steps, upon effective reduction to practice, not providing any guarantee of success in the succeeding steps.

Level of Skill in the Art

The level of skill in the art is deemed to be high.

Conclusion

Thus given the broad claims in an art whose nature is identified as unpredictable, the unpredictability of that art, the large quantity of research required to define these unpredictable variables, the lack of guidance provided in the specification, the absence of a working example and the negative teachings in the prior art balanced only against the high skill level in the art, it is the position of the examiner that it would require undue experimentation for one of skill in the art to perform the method of the claim as broadly written.

Response to Arguments

The response traverses the rejection. The response asserts the cancer and samples were amended and thus the rejection is overcome. This argument has been considered but is not convincing because the art specifically teaches pancreatic cancer shows a downregulation of SPSB. In particular Habermann teaches SPSB2 is downregulated compared to healthy controls in **pancreatic adenomas** (Table 1). Thus, it is unpredictable if and when the skilled artisan could use SPSB2 protein/gene expression to make treatment decisions for pancreatic cancer.

The response further states that Claim 17 excludes patients who have a urinary tract infection. This claim specifically exemplifies that Claim 13 encompasses patients who have a urinary tract infection. The specification clearly provides that it is

unpredictable this method may be performed with patients who have urinary tract infections. The specification teaches urinary tract infections also show expression of SPSB2 in patients with urinary tract infection is higher than bladder cancer patients (para 75). It would require further unpredictable and undue experimentation to determine how to diagnose cancer in patients with urinary tract infections using SPSB2 expression. This would require significant inventive effort, with each of the many intervening steps, upon effective reduction to practice, not providing any guarantee of success in the succeeding steps.

Thus for the reasons above and those already of record, the rejection is maintained.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

8. Claim(s) 13, 15, 18 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Habermann et al. (US 2013/0196876, August 1, 2013).

Habermann teaches analyzing SPSB2 protein by incubating a sample with antibodies with a protein encoded by SPSB2 (see para 10) and comparing the amount

to a control. Habermann teaches SPSB2 is downregulated compared to healthy controls.

With respect to Claim 15, Habermann teaches using blood or sera samples (para 39).

With respect to Claim 18, no treatment is administered.

Response to Arguments

The response traverses the rejection. The response asserts Habermann does not teach the sample is tissue or urine. This argument has been considered but is not convincing because Habermann teaches the sample may be blood or serum. Blood is a connective tissue. Thus for the reasons above and those already of record, the rejection is maintained.

Conclusion

9. No claims allowable.

10. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Zeng et al. (Gastroenterology, Vol. 150, pages 1633-1645, 2016) teaches SPSB2 has a SNP that is associated with colorectal risk (see Table 1).

Ping et al. (Carcinogenesis, Vol. 41, No. 7, pages 887-893, 2020) teaches analyzing differences in gene-expression profiles in breast cancer. Ping teaches analyzing SPSB2 (see Table 2, halfway down). The claim encompasses not performing any surgery or treatment to subject who were not affected by cancer. Figure 1 illustrates normal patients were treated. These patients were not treated.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

11. Any inquiry concerning this communication or earlier communications from the examiner should be directed to JEANINE ANNE GOLDBERG whose telephone number is (571)272-0743. The examiner can normally be reached Monday-Friday 6am-3:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wu-Cheng (Winston) Shen can be reached on (571) 272-3157. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/JEANINE A GOLDBERG/
Primary Examiner, Art Unit 1682
May 26, 2026